FA Activity Report: Ocean Waste Detection using Image Processing

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Abstract

This report presents a computer vision–based system for detecting ocean waste using image processing techniques. By leveraging deep learning models and high-resolution imagery, the system enables automated classification of marine debris, supporting environmental monitoring and cleanup operations. The proposed approach integrates image preprocessing, feature extraction, and CNN-based classification.

Keywords — Ocean Waste, Image Processing, Deep Learning, Marine Debris Detection

I. AIM

To develop an image processing system that automatically detects and classifies ocean waste, enabling scalable monitoring and supporting marine conservation efforts.

II. OBJECTIVES

1. To collect ocean images from drones, satellites, and underwater cameras.

2. To preprocess and enhance images for better waste detection.

3. To apply deep learning models for detecting and classifying different types of waste.

4. To build datasets that can support further research and policymaking. 5. To support real-time monitoring and alerts for cleanup operations.

III. MOTIVATION

Marine pollution is a critical global challenge, with millions of tons of plastic entering oceans annually. Traditional waste monitoring is slow, labor-intensive, and costly. Automated detection using image processing can scale across large regions, providing accurate and timely insights. This project highlights how AI can aid environmental sustainability.

IV. LITERATURE SURVEY

Title	Author & Year	Work Presented	Methodol og y	Perf. Measures	Key Findings
Deep Learning for Marine Debris Detection	Martinez-Vi ce nte et al., 2022	Detecting floating plastics in ocean images	CNN-bas ed classificati on	Precision, Recall	High accuracy in detecting plastics from drone imagery
Automatic Plastic Waste Detection in Waterways	Kikaki , 2021	Marine litter recognition	Object detection (YOLOv3)	mAP, IoU	Achieved robust detection under varying light conditions
UAV-based Plastic Waste Mapping	Topouzelis, 2019	Drone monitoring of coastal waste	Image segmentation	Accuracy, F1-score	Effective for coastal cleanup planning
Satellite Detection of Floating Debris	Biermann, 2020	Sentinel-2 satellite data for plastics	Spectral image analysis	Classificati on accuracy	Detected plastic patches from orbit
Computer Vision for Ocean Cleanup	Maximen ko, 2021	Vision-bas ed debris monitoring	Hybrid ML & vision models	ROC curve	Improved detection of fishing nets & large debris
Underwater Image Processing for Waste	Li ,2018	Detecting submerged waste	Preproces sin g + CNNs	PSNR, Recall	Enhanced detection in turbid water

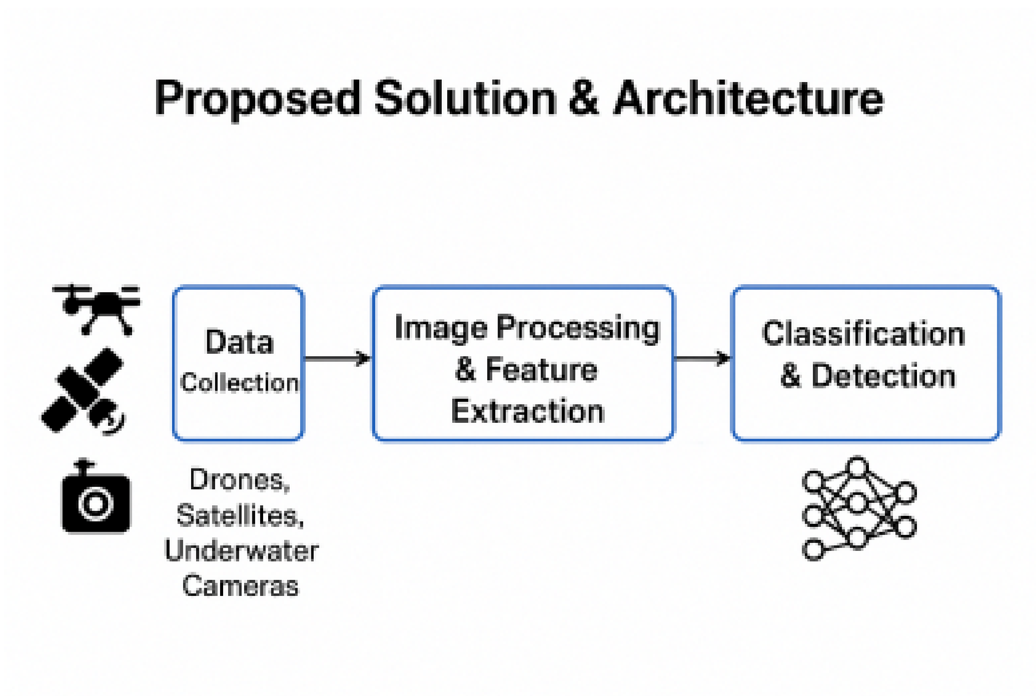
Detection					
Marine Pollution Monitoring Using AI	Chen, 2023	AI framework for debris detection	Transfer learning on ocean datasets	Accuracy, Loss	Outperfor me d baseline ML models in classificatio n

V. PROPOSED SOLUTION & ARCHITECTURE

The proposed solution integrates three main stages:

1. **Data Collection** — Acquire images from drones, satellites, and underwater sensors.
2. **Image Processing & Feature Extraction** — Apply noise reduction, segmentation, and edge detection.
3. **Classification & Detection** — Use deep learning models like CNNs and YOLO for waste classification.

The system can be extended into a web application, allowing real-time uploads and monitoring dashboards for NGOs and policymakers.



VI. METHODOLOGY

1. Data Collection: Gather datasets from open repositories and drone missions.

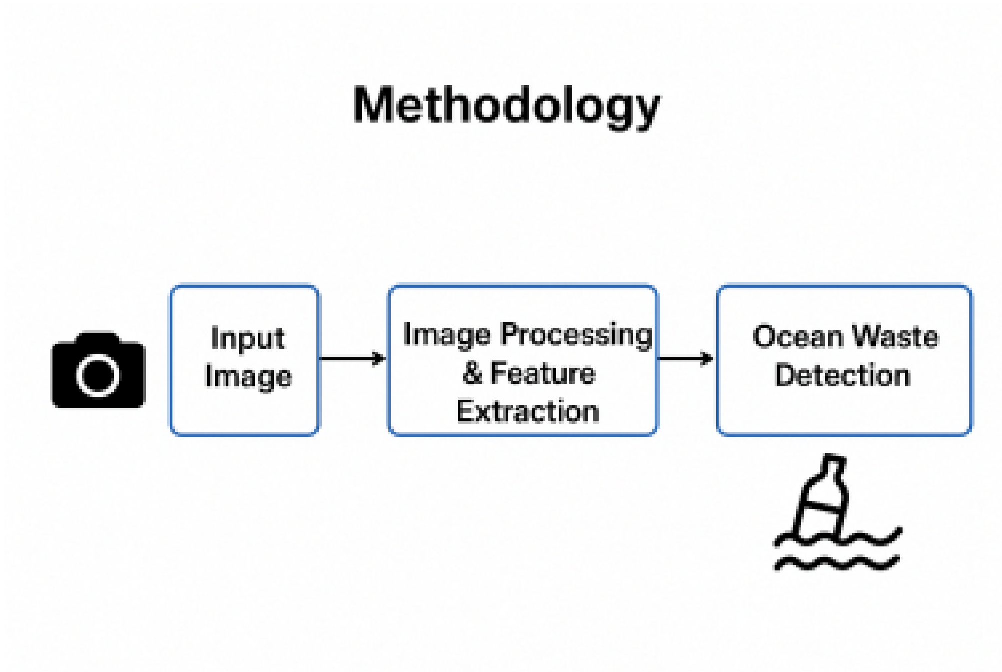
2. Preprocessing: Apply histogram equalization, denoising, and image resizing.

3. Feature Engineering: Extract color, shape, and texture features. 4. Modeling:

Train CNN/YOLO models on annotated datasets.

5. Validation: Evaluate with precision, recall, F1-score.

6. Deployment: Provide visualization and real-time monitoring through dashboards.



VII. CONCLUSION

Ocean waste poses a severe threat to marine ecosystems. This report demonstrates how image processing and AI can provide scalable, automated, and cost-effective waste detection solutions. By leveraging drones, satellites, and deep learning, monitoring can be transformed into an efficient system that aids conservation and cleanup operations.

VIII. REFERENCES

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